MIL-S-55389(EL)
30 March 1973

MILITARY SPECIFICATION

SUPPRESSOR, ELECTRICAL TRANSIENT, MX-7778A/GRC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a shunt-type transient suppressor assembly designated as Suppressor, Electrical Transient MX-7778A/GRC. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871	Twine, Cotton, Wrapping
NN-P-71	Pallets, Material Handling, Wood, Double Faced, Stringer Construction
QQ-S-781	Strapping, Flat, Steel
PPP-B-20	Bag, Cotton, Mailing
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-T-76	Tape; Pressure Sensitive Adhesive Paper, (For Carton Sealing)
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced

FSC 5915

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the currentversion before use.

MILITARY

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-V-173	Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40 Inch x 48 Inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems

DRAWINGS

ELECTRONICS COMMAND

DL-SM-B-745200	Drawing List of Suppressor, Electrical Transient, MX-7778( )/GRC

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:19Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First Article. - When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. - The equipment shall be constructed in accordance with Drawing DL-SM-B-745200.

3.3 Bounce, Preconditioning. - The equipment shall meet the requirements herein, without subsequent processing, after subjection to the bounce preconditioning specified in 4.5.4.

3.4 Electrical Requirements.

3.4.1 Power. - The equipment shall operate with tactical radios operating from a 28 volt 100 to 400 amp vehicular power supply.

3.4.2 Voltage limitation. - With the output of the transient suppressor open circuited and the input voltage adjusted such that the zener diode draws 1 ± 0.1 amperes at two (2) seconds of this voltage application the maximum allowable output voltage shall be within a range of from 33.0 to 36.0 volts referred to a zener case temperature of 75 degrees F. The unit shall be tested normally with a zener case temperature of 75 degrees F. When test requirements are at temperatures other than 75 degrees F, subtract 17.2 mv per degree F from the output voltage for a lower zener case temperature and add 17.2 mv per degree F to the output voltage for a higher zener case temperature. (See 4.7.1)

3.4.3 Circuit breaker operation.

3.4.3.1 Load, Electrical. - With the output load drawing 70 ± 0.5 amperes, the suppressor circuit breaker shall trip within 5 seconds in a 77 ± 5 degrees F. environment. The unit shall be tested normally in a 77 ± 5 degree F. environment. When test requirements are at a -40 degree F. environment, the circuit breaker shall trip within 30 seconds. When test requirements are at + 155 degree F. environment, the circuit breaker shall trip within 5 seconds. (See 4.7.2.1)

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

3.4.3.2 No Load, (Thermal). - With the output of the suppressor open-circuited and the input voltage adjusted such that the zener diode draws a current of 20 ± 0.5 amperes, the circuit breaker shall trip within 5 minutes. The unit shall be allowed to temperature stabilize for 5 minutes before further testing. (See 4.7.2.2)

3.4.4 Pulsing. - The suppressor shall operate, without the circuit breaker tripping, when pulsed five successive times at ten-second intervals, with a 100 + 0, -5V, 50 millisecond pulse (as measured without the suppressor connected) from a power supply having a 0.50 + 0, - .05 ohms internal impedance and capable of supplying 200 amperes for 50 milliseconds. Series connected storage batteries having a total internal and external resistance of 0.50 + .00 ohms as
- .05
measured during the pulse and ahead of the suppressor can be used. The rise in output voltage of the suppressor during each pulse measured from 1 millisecond after the start of the pulse to the end of pulse shall not exceed 7.0 volts and the current droop shall not exceed 20.0 amperes. The assembly shall meet the requirements of 3.4.2 after pulsing. (See 4.7.3) The output voltage during each pulse shall not fall below the voltage as measured in 3.4.2.

3.4.4.1 Pulse Circuit. - The pulsing circuit shall be capable of delivering five successive 50 millisecond pulses at ten second intervals of 125 amperes minimum when loaded with a 0.30 + 0.00
- 0.05
ohms resistive load. The pulse rise time shall be not longer than one millisecond. The pulse droop shall be less than 10 amperes at 50 milliseconds. The total internal and external impedance of the voltage source and pulser shall be 0.50 + 0.00 ohms. (See 4.7.3.1)
- 0.05

3.4.5 Voltage Drop and Battle Override. - The voltage drop measured at the positive and negative terminals of both mating connectors to the test item shall not exceed 0.40 volts with a 45 + 0 - 0.5 ampere load. The voltage drop shall be measured, first, with the circuit breaker "on" and the battle override switch "off", and, second, with the circuit breaker "off" and the battle override switch "on". The input to the test item shall not exceed 30.0 volts during the test. (See 4.7.4)

3.4.6 Cable Assembly. -

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:19Z
Check the source to verify that this is the current version before use.

3.4.6.1 Continuity (ohmeter). - Verify continuity between terminal A of connector (male) and terminals A and B of connector (female). Verify continuity between terminal B of connector (male) and terminals C and D of connector (female). (See 4.7.5)

3.4.6.2 Voltage Drop. - The total voltage drop measured at the positive and negative terminals of both mating connectors to the cable assembly shall not exceed 1.0 volt with a 45 + 0 - 0.5 ampere load. (See 4.7.5)

3.4.7 Self Protection. - With the output of the suppressor open-circuited, the input voltage shall be adjusted such that the zener diode will draw a current of 45 + 5, -0 amperes when the suppressor circuit breaker is closed. The circuit breaker shall then be closed and allowed to trip to protect the zener diode. After at least one (1) minute waiting period, again with the output open-circuited; the input voltage shall be adjusted such that zener diode will draw a current of 55 + 5, -0 amperes when the suppressor circuit breaker is closed. The circuit breaker shall be closed again and allowed to trip. The suppressor shall then be allowed to temperature stabilize for five (5) minutes after which the test specified in 3.4.2 shall be conducted to verify that the suppressor was not damaged. (See 4.7.6)

3.4.8 Pulse Cycling. - The circuit breaker shall not trip when the suppressor is subjected to the test sequence described in 4.7.7.

3.5 Service Conditions. - The equipment shall meet the following service conditions. The unit shall meet the electrical requirements of 3.4.2, 3.4.3.1, 3.4.3.2, 3.4.4 and 3.4.5 in those service conditions requiring unit operation. Continuous use for a period of 5 years, under the world-wide environmental conditions specified in 3.5.1 through 3.5.10 with a duty cycle of 8 hours on and 4 hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.7. Approval of the First Article Inspection samples (see 3.1) shall be considered as compliance with this requirement. (See 4.6)

3.5.1 Temperature. -

3.5.1.1 High Temperature. - The unit shall not be damaged when subjected to a storage temperature of + 160°F. The unit shall meet the electrical requirements specified in 3.5 when operating at an ambient temperature of + 155°F, with the exception of 3.4.4 which shall be conducted using a 90 +0 -5 volt, 50 millisecond pulse. The circuit breaker shall not trip when the unit is subjected to a load current of 45 amperes for 90 minutes when operating at an ambient temperature of 155°F. (See 4.6.1.1)

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

3.5.1.2 Low Temperature. - The unit shall not be damaged when subjected to a storage temperature of -65°F. The unit shall meet the electrical requirements specified in 3.5 when operating at an ambient temperature of -40°F. (See 4.6.1.2)

3.5.2 Humidity (Moisture Resistance). - During testing as specified in 4.6.2 the equipment shall meet the electrical requirements of 3.5.

3.5.3 Altitude. - The equipment shall not be damaged when subjected to an altitude of 50,000 feet above sea level. The equipment shall meet the electrical requirements specified in 3.5 when subjected to an altitude of 10,000 feet above sea level. (See 4.6.3)

3.5.4 Vibration. -

3.5.4.1 Resonance and Vibration Cycling. - When tested in accordance with 4.6.4.1 the equipment shall meet the requirements of 3.4.2 during vibration and the equipment shall meet the electrical requirements specified in 3.5, following the vibration tests and the equipment shall show no mechanical or electrical damage which could prevent it from meeting operational service or maintenance requirements.

3.5.4.2 Bounce. - The equipment shall show no more than surface abrasion when tested in accordance with 4.6.4.2. The equipment shall meet the electrical requirements of 3.5 after meeting the bounce requirement.

3.5.5 Shock. -

3.5.5.1 Basic Design. - The equipment shall meet the electrical requirements specified in 3.5 following the test in accordance with 4.6.5.1.

3.5.5.2 Crash Safety. - The equipment shall show no failure of its attachment and shall remain in place and not create a hazard when tested in accordance with 4.6.5.2. However, bending and distortion shall be permitted. The equipment shall meet the electrical requirements specified in 3.5 after meeting the crash safety requirement.

3.5.6 Sand and Dust. - The equipment (including accessories) shall show no mechanical or electrical damage when tested in accordance with 4.6.6. Following the tests of 4.6.6 the unit shall meet the electrical requirements of 3.5.

3.5.7 Immersion. - The equipment (including accessories) shall be capable of being immersed as described in 4.6.7 without any evidence of water leakage.

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

3.5.8 Orientation. - The equipment shall be tested in accordance with 4.6.8 at an orientation of 20 degrees from normal operating position. During each orientation the unit shall meet the requirements of 3.4.3.1, 3.4.3.2 and 3.4.4.

3.5.9 Salt Fog. - When subjected to salt atmosphere described in 4.6.9 the unit (including accessories) shall show no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface.

3.5.10 Fungus. - The equipment, (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.6.10. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surface. When overall tropicalization treatment of equipment is a contractual requirement, a moisture and fungus resistant coating of varnish conforming to MIL-V-173 shall be applied in accordance with MIL-T-152. The moisture-fungus resistant coating shall be applied after the equipment has been cleaned to remove such contaminants as corrosion products, dust fingerprints, lubricating oils, mold release agents, sand and solder fluxes. When tape is used on parts of equipment that are masked against treatment, the tape shall be an electrical grade. In the application of the MFP coating, care shall be taken to assure that the coating does not affect circuit constants and thereby degrade performance.

3.6 Parts, Materials, and Processes; General. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements, therein, shall apply. (See 3.2)

3.7 Finish, protective. - Equipment shall be given protective finish in accordance with specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on Type I surfaces shall be green color (olive drab), semigloss enamel. (See 4.4)

3.7.1 Color. - The final paint film on Type I surfaces shall be smooth semigloss, green, color conforming to Nr. X24087 of Federal Standard FED-STD-595. (See 4.4)

3.8 Marking. - Marking shall conform to Specification MIL-M-13231. Panel marking shall be Group I as described in that specification. (See 4.4)

3.8.1 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment. (See 4.4)

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

3.8.2 Serial numbers. - Each unit, MX-7777( )/GRC shall be serial numbered. (See 4.4)

3.9 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.8)

3.10 Technical literature, tools and running spare parts. - Technical literature, tools and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on order. (See 6.2g & 6.2h)

3.11 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268:

(1) General requirements for plastic material and plastic parts.

(2) Wiring and cabling, including: Slack, Protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

(3) Flux and cleaning agents for solder.

(4) Soldering.

(5) Cleaning.

(6) Riveting.

(7) General requirements for securing of parts.

(8) Welding.

(9) Tropicalization.

(10) Mechanical operation of controls.

(11) Self-tapping screws.

(b) In this Specification:

(1) Marking.

(2) Finish.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T23:19Z
Check the source to verify that this is the current version before use.

3.12 Safety. - The equipment shall meet the pertinent safety requirements of requirement 1 of MIL-STD-454.

3.13 Human Factors. - The equipment shall meet the pertinent requirements of MIL-STD-1472.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection requirements specified herein are classified as follows:

(a) First Article Inspection Does not include preparation for delivery. (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection:

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.11)

4.3 First Article Inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified in Group A, Group B. (See Tables II and III.) First Article Inspection shall be performed on the samples specified and in the sequence shown in Table I. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:19Z
Check the source to verify that this is the current version before use.

TABLE I FIRST ARTICLE INSPECTION

INSPECTIONS	REQT. PARA.	TEST PARA.	UNIT 1	UNIT 2	UNIT 3
Inspections covered by subsidiary documents		4.4	Inspections to be performed on all units		
Bounce Preconditioning	3.3	4.5.4	Inspections to be performed on all units		
GROUP A Inspection	(See Table II)		Inspection to be performed on all units		
GROUP B Inspection	(See Table III)		Inspection to be performed on all units		
Environmental					
Low Temperature	3.5.1.2	4.6.1.2	1		
High Temperature	3.5.1.1	4.6.1.1	2		
Orientation	3.5.8	4.6.8	3		
Relative Humidity	3.5.2	4.6.2	4		
Altitude	3.5.3	4.6.3		1	
Salt Fog	3.5.9	4.6.9		2	
Immersion	3.5.7	4.6.7			1
Sand-Dust	3.5.6	4.6.6			2
Vibration, Resonant Search	3.5.4.1	4.6.4.1			3
Vibration, Bounce	3.5.4.2	4.6.4.2			4
Shock, Ballistic	3.5.5	4.6.5			5
Fungus	3.5.10	4.6.10			6
Pulse Circuit	3.4.4.1	4.7.3.1			Test Equipment

Source: https://assist.dla.mil -- Downloaded: 202640-22T23:19Z
Check the source to verify that this is the current version before use.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where Required
Parts, Materials & Processes	3.6
Finish	3.7
Color	3.7.1
Marking	3.8
MFP Treatment	3.5.10
Tech Lit, Tool & Running spare parts	3.10
Safety	3.12
Human Factors	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspection specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:19Z
Check the source to verify that this is the current version before use.

TABLE II - GROUP A INSPECTION

Inspection	Req. Par.	Insp. Par.	AQL Major	AQL Minor
Visual & Mechanical	3.11	4.9	1%	4.0%
Voltage Limitation	3.4.2	4.7.1	1%	*
Load, Electrical	3.4.3.1	4.7.2.1	For the	*
No Load, Thermal	3.4.3.2	4.7.2.2	Group	*
Pulsing	3.4.4	4.7.3		*
Cable Assembly	3.4.6	4.7.5		*
Self Protection	3.4.7	4.7.6		*

* All electrical defects are considered major.

4.5.1.1 Order of Inspection within Group A. - Group A inspection shall be performed in any order satisfactory to the Government.

4.5.2 Group B Inspection. - Group B inspection shall normally be performed in inspection lots that have passed Group A inspection and on samples that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

TABLE III - GROUP B INSPECTION

Inspection	Req. Par.	Insp. Par.	AQL
Voltage Drop & Battle Override	3.4.5	4.7.4	1.0%
Pulse Cycling	3.4.8	4.7.7	1.0%
Interchangeability	3.9	4.8	4.0%

4.5.3 Group C Inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

TABLE IV - GROUP C INSPECTION

Inspection	Req. Para.	Insp. Para.
Group I		
Immersion	3.5.7	4.6.7
Group II A		
High Temperature	3.5.1.1	4.6.1.1
Low Temperature	3.5.1.2	4.6.1.2
Altitude	3.5.3	4.6.3
Salt Fog	3.5.9	4.6.9
Group II B		
Vibration, Resonant Search	3.5.4.1	4.6.4.1
Vibration, Bounce	3.5.4.2	4.6.4.2
Shock, Ballistic	3.5.5	4.6.5
Group III A		
Orientation	3.5.8	4.6.8
Relative Humidity	3.5.2	4.6.2
Group III B		
Sand and Dust	3.5.6	4.6.6
Fungus	3.5.10	4.6.10
Group IV		
Pulse Circuit Test	3.4.4.1	4.7.3.1

4.5.3.1 Sampling for Group C Inspection. -

4.5.3.1.1 Group I, Table IV. - Two samples shall be selected at random, from each 100 equipments produced and subjected to each inspection specified. The first two samples shall be selected at the start of the contract from the first quality conformance lot. (See 4.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:19Z
Check the source to verify that this is the current version before use.

4.5.3.1.2 Groups II A and II B. - Two samples shall be selected at random for each sub-group, from each 500 equipments produced and subjected to each inspection specified. The first two samples shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5)

4.5.3.1.3 Groups III A and III B. - One sample shall be selected at random for each sub-group, from each 1000 equipments produced and subjected to the inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5)

4.5.3.1.3.1 Group IV. - The pulse circuit test shall be performed once for each 1000 equipments produced.

4.5.3.1.4 Reinspection of conforming Group B and Group C Sample Units. - Unless otherwise specified, sample units which have been subjected to and passed group B and group C inspection, may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.3.2 Group C Failures. - Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Bounce Preconditioning. - Each unit of each lot shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection. (See 3.3)

4.6 Service condition tests. - The following tests constitute service condition tests:

Voltage Limitation	4.7.1
Load, Electrical	4.7.2.1
No Load, Thermal	4.7.2.2
Pulsing	4.7.3
Voltage Drop and Battle Override	4.7.4

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

4.6.1 Temperature tests. -

4.6.1.1 High Temperature. - The equipment shall be subjected to Test Method 501, Procedure I of MIL-STD-810. The storage temperature shall be + 160°F and the operating temperature shall be + 155°F. The unit shall meet the requirements of 3.5.1.1 at the + 155°F. temperature. In Step 7 of the Procedure the unit shall be subjected to the tests of 4.6.

4.6.1.2 Low Temperature. - The equipment shall be subjected to the test of Method 502, Procedure I of MIL-STD-810. The storage temperature ambient shall be - 65°F. The operating temperature ambient shall be - 40°F. During operating conditions the unit shall be subjected to the tests of 4.6. (See 3.5.1.2)

4.6.2 Humidity (moisture-resistance) test. - The equipment (including accessories) shall be tested in accordance with Method 507, Procedure III, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 4.6. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. (See 3.5.2)

4.6.3 Altitude test. - The equipment shall be tested in accordance with Method 500, Procedure I of MIL-STD-810. During the operating cycle the equipment shall be subjected to the tests of 4.6. (See 3.5.3)

4.6.4 Vibration tests. -

4.6.4.1 Resonance and Vibration Cycling. - The equipment shall be subjected to the test of Method of 514.1, Procedure VIII of MIL-STD-810. Curve W of Figure 514.1-6 shall be used in conjunction with Table 514.1VI for a mileage schedule of 6000 miles. The unit shall meet the requirements of 3.5.4.1.

4.6.4.2 Bounce. - The equipment shall be subjected to the test of Method 514, Procedure IX, Part 2 of MIL-STD-810. (See 3.5.4.2)

4.6.5 Shock. -

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

4.6.5.1 Basic Design. - The equipment shall be subjected to the test of Method 516.1 Procedure 1 of MIL-STD-810. The shock pulse shall be in accordance with Figure 516.1-1 using shock parameters b and d of that figure. The equipment shall be subjected to the tests of 4.6 following the test.

4.6.5.2 Crash Safety. - The equipment shall be subjected to the test of Method 516.1-1, Procedure III of MIL-STD-810. The shock pulse shall be in accordance with Figure 516.1-1 using shock parameters b and d of that figure. The equipment shall meet the requirements of 3.5.5.2.

4.6.6 Sand and dust test. - The equipment (including accessories) shall be subjected to the test of Method 510, Procedure I of MIL-STD-810. The equipment shall be subjected to the tests of 4.6 following the tests. (See 3.5.6)

4.6.7 Immersion test. - The equipment (including accessories) shall be subjected to the test of Method 512, Procedure I of MIL-STD-810. (See 3.5.7)

4.6.8 Orientation test. - The equipment shall be inclined for a minimum of 1 minute in each plane (forward, backward, left and right) to an angle of 20 degrees. During inclination in each plane, the equipment shall be subjected to the tests of 4.7.2.1, 4.7.2.2 and 4.7.3. (See 3.5.8)

4.6.9 Salt Fog. - The equipment (including accessories) shall be subjected to the test of Method 509 of MIL-STD-810. The unit shall meet the requirements of 3.5.9.

4.6.10 Fungus. - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open-condition during the test. Operation is not required. (See 3.5.10)

4.7 Electrical tests. -

4.7.1 Voltage Limitation test. - Using standard laboratory test instruments the transient suppressor shall be tested for conformance to the requirements of 3.4.2.

4.7.2 Circuit breaker operation test. -

4.7.2.1 Load, Electrical. - Using standard laboratory test instruments the transient suppressor shall be tested for conformance to the requirements of 3.4.3.1.

Source: https://assist.dla.mil -- Downloaded: 202640-22T23:19Z
Check the source to verify that this is the currentversion before use.

4.7.2.2 No Load; Thermal. - Using standard laboratory test instruments the transient suppressor shall be tested for conformance to the requirements of 3.4.3.2.

4.7.3 Pulse Test. - Using standard laboratory test instruments the transient suppressor shall be tested for conformance to the requirements of 3.4.4.

4.7.3.1 Pulse Circuit Test. - The pulse circuit shall be tested to assure conformance to the requirements of 3.4.4.1. The waveform recordings shall be made on a Tektronix Type No. 549 Oscilloscope or equivalent.

4.7.4 Voltage Drop and Battle Override Test. - The voltage drop shall be measured by adding the voltage drop from positive input to positive output and the voltage drop from negative input to negative output. The test shall be duplicated to incorporate both output connectors. Using standard laboratory test instruments the transient suppressor shall be tested for conformance to the requirements of 3.4.5.

4.7.5 Cable Assembly Test. - Using standard laboratory test instruments the transient suppressor shall be tested for conformance to the requirements of 3.4.6.

4.7.6 Self Protection Test. - Using standard laboratory test instruments the transient suppressor shall be tested for conformance to the requirements of 3.4.7.

4.7.7 Pulse Cycling Test. - Using standard laboratory instruments and test equipment, the transient suppressor shall be tested for conformance to the requirements of 3.4.8, when the unit is subjected to an ambient of + 155°F and the following test sequence:

Step I	- Apply 50 amperes minimum for five. (5) 10% -0 minutes.
Step II	- Immediately decrease current to 25 +2% -0 amperes and hold for one (1) ±10% seconds.
Step III	- Immediately increase current to 100 +2% -0 amperes and hold for thirty (30) + 10% -0 milliseconds.
Step IV	- Immediately decrease current to 50 amperes minimum and hold for ten (10) +10% -0 seconds.

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

Step V	- Immediately decrease current to 25 +2% -0 amperes and hold for one (1) ±10% second.
Step VI	- Immediately increase current to 100 +2% -0 amperes and hold for thirty (30) + 10% -0 milliseconds.
Step VII	- Repeat Steps 4, 5, & 6 eight (8) times.

In the steps above the input voltage to the suppressor shall not exceed 30 Volts.

4.8 Interchangeability. - The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirements of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.9 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in MIL-STD-252.

4.10 Rough handling test (preparation for delivery). - When the rough handling test in accordance with MIL-P-116 is required by contract (see 6.2(d)(2)(c)) the unit shall be subjected to the tests of 4.7.1 thru 4.7.7 to determine freedom from malfunction cuased by the rough handling.

4.11 Quality Conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:19Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C, as specified.

5.1.1 Level A.

5.1.1.1 Cleaning. - Suppressor, Electrical, Transient MX-7778A/GRC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Suppressor, Electrical, Transient MX-7778A/GRC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Cable Assembly, CX-10613/G. - Each cable assembly shall be packaged method 111. Coil each cable into a coil of proportionate dimensions and tie in three places with cotton twine conforming to T-T-871. Secure the connectors to the coil with the ties.

5.1.1.4.3 Template. - The template shall be packaged method 1C-1.

5.1.1.4.4 Mounts and Nuts. - Items bearing the same stock number shall be packaged method 111 by Placing within a draw-string bag, conforming to PPP-B-20, close the bag with string ties.

5.1.1.4.5 MX-7778A/GRC. - MX-7778A/GRC shall be packaged method 111. Cushion the unit, on all surfaces with cells or pads both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned unit and the items, packaged as specified in 5.1.1.4.2 through 5.1.1.4.4, within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. - Suppressor, Electrical, Transient MX-7778A/GRC, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be Level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2(b))

5.2.1 Level A.

5.2.1.1 Consolidation. - A quantity of Suppressor, Electrical, Transient MX-7778A/GRC, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized Load. - A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, class B.

5.2.2 Level B.

Source: https://assist.dla.mil -- Downloaded: 202640-22T23:19Z
Check the source to verify that this is the current version before use.

5.2.2.1 Consolidation. - A quantity of Suppressor, Electrical, Transient MX-7778A/GRC, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized Load. - A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C.

5.2.3.1 Consolidation. - A quantity of Suppressor, Electrical, Transient MX-7778A/GRC, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers, packed as specified in 5.2.3.1, shall be used as the shipping container. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - The transient suppressor is intended to be used with tactical radios (AN/VRC-12, AN/VRC-53, AN/VRC-106, and their configurations when operated from vehicular power supplies) in all military vehicles.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of preservation and packaging and packing. (See Section 5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:19Z
Check the source to verify that this is the current version before use.

(c) The specific paragraphs of Section 5 which are applicable to this procurement.

(d) Preproduction inspection:

(1) Three sample units of each item cited in the section 1 are generally required so that lengthy and potentially damaging environmental tests can be performed on several sample units. (See 3.1)

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.10)

(h) Quantity of tools and running spare parts required. (See 3.10)

(i) Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer. (See 3.8.3)

6.3 Color. - The color chip furnished by the procuring agency (See 3.7.1) will be color chip no. X-24087, and may be obtained upon request to Commanding General, Coating and Chemical Laboratory, Aberdeen Proving Grounds, Maryland.

6.4 Group C Inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3)

6.5 Location of inspection. - It is desirable that the inspection be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.

6.6 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Qualitative material requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for the contractor where he is required to make a selection of parts, materials, processes, constructive methods, etc.

6.7.1 Temperature. -

(a) Operating: Ambient temperature in the range of + 155°F to - 40°F. (The temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Non-operating: Exposure in the range of + 160°F to - 80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

6.7.2 Relative humidity. - Exposure to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours for each 24 hour period.

6.7.3 Altitude (elevation). -

(a) Operating: Up to 10,000 feet above sea level.

(b) Non-operating: Up to 50,000 feet above sea level.

6.7.4 Orientation. -

(a) Operating. - Any orientation up to 20 degrees from normal operating position (that is: forward, backward, left, or right).

(b) Non-operating: Storage in any position for a period of two years.

CUSTODIAN:

Army-EL

PREPARING ACTIVITY

ARMY-EL

PROJECT NR. 5915-A037

Source: https://assist.dla.mil -- Downloaded: 202610-22T23:19Z
Check the source to verify that this is the current version before use.